Name: _____

Section: _____

Instructions: Show supporting work to receive full credit for each problem. No calculators, notes, or other external resources are allowed. The quiz will be graded out of 10 points.

1. Plot the phase portrait of the autonomous ODE given by

$$\frac{dy}{dx} = (1-y)^2(y-3)(y+2).$$

Classify each equilibrium solution as stable, unstable, or semi-stable.

2. Is the equation $\frac{dy}{dx} = ye^{-x^2+2y}$ separable? Explain why or why not.